

Segment -01

15/03/2026

Microprocessor

★ What is a Computer system?



A computer is a programmable machine that:

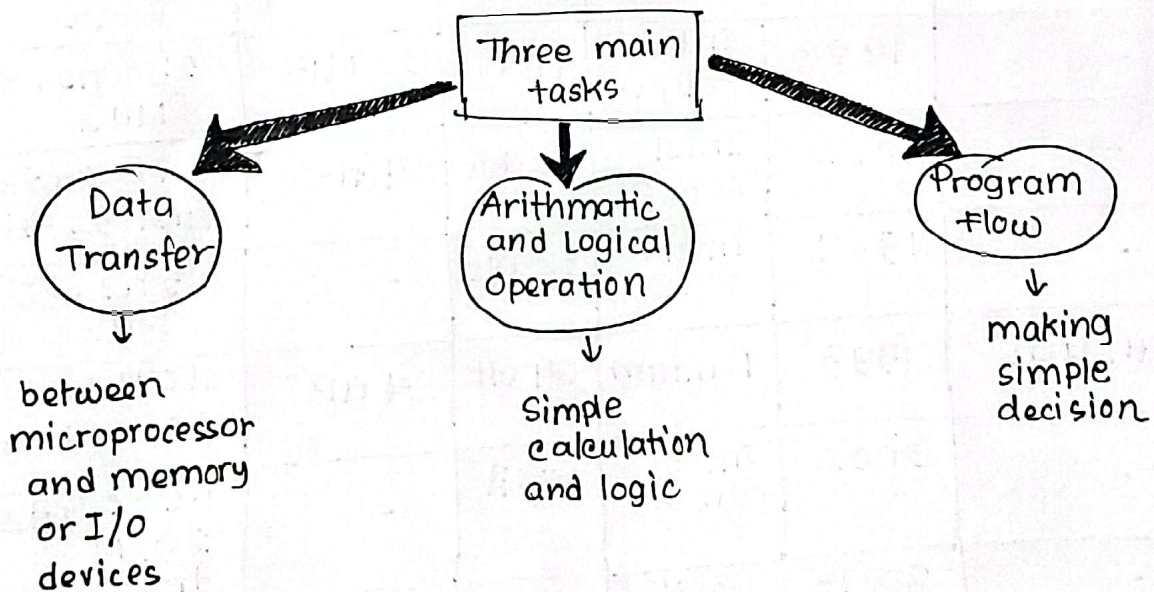
- Receives input
- Stores and manipulates data/infos
- Provides output in a useful format

★ Microprocessor:

Heart of
MP based
computer is:
IC
(Integrated
Circuit)

Controlling
element
in
computer

Controls
memory and
I/O devices
through
buses



→ can execute billions of instructions per second from a program stored in memory.

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★ Evaluation of Microprocessor%

Generation	Year	Model	Bits	Memory	Speed/Notes
1st	1971	Intel 4004/4040	4-bit	4-bit	50 KIPs, P-channel MOSFET
2nd	1971	Intel 8008	8-bit	16KB	48 instructions, 20 μ sec exec. time
3rd (1st modern)	1973	Intel 8080	8-bit	64 KB	TTL technology, 2 μ s exec. time
—	1977	Intel 8085	8-bit	64 KB	1.3 μ sec, Internal clock 3 MHz
—	1978	Intel 8086	16-bit	1 MB	400 nano-sec, 5 MHz, 6 byte cache
—	1979	Intel 8088	16-bit	—	Faster than 8086
—	1983	Intel 80286	16-bit	16 MB	250 ns, 8 MHz
—	1986	Intel 80386	32-bit	4 GB	8 KB cache, first 32-bit
—	1989	Intel 80486	32-bit	—	25 ns, 50 MHz
5th Gen	1993	Pentium	64 bit	4 GB	110 MIPs, 60-66 MHz
—	2006	Intel core 2	64 bit	—	1.2-3 GHz, 291 M transistors
—	2007-2010	i3, i5, i7	64 bit	—	2.2-3.6 GHz

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★ Microcontroller:

- A single chip microcomputer that includes:

- Microprocessor
- 64 byte RAM
- 1-2 KB ROM
- Several signal lines for I/O

- Fabricated using VLSI Technology
- Used in: appliances, traffic lights etc
- Example: Zilog-28, Intel MC551, Motorola 68HC11

Micro processor VS Microcontroller

Feature	Microprocessor	Microcontroller
Heart of	Computer system	Embedded system
Memory I/O	External	Built-in
Circuit size	Large	Small
Compact use	x NO	✓ Yes
Cost	High	Low

★ Important Terms:

Term	Defination
Bit (b)	Binary digit - 0 or 1
Byte (B)	8 bits
Word (W)	2 bytes = 16 bits
Instruction	Binary command executed by a computer

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★ Types of Microprocessor:

① CISC — Complex Instruction Set Computer:

- Single instruction can do multiple low level operations (load, store, arithmetic operation)
- Ex: Intel 386, Intel 486, Pentium

ADD A, B

- Load A from memory
- Add B to it
- Store the result back in memory.

② RISC — Reduced Instruction set computer:

- CPU uses simple, fast instructions, each executing in one clock cycle
- tasks broken into small steps.

↓
So that processor can execute them faster and more efficiently

LOAD R1, A

LOAD R2, B

ADD R3, R1, R2

STORE R3, C

③ EPIC — Explicitly Parallel Instruction Computing

- compiler groups independent instructions for parallel execution
- No need for higher clock frequencies.

analyze program

groups independent instruction together

executed at the same time by CPU

Suppose,

- $A + B$
- $C + D$
- $E + F$

$\{ \text{ADD } A, B \} \{ \text{ADD } C, D \} \{ \text{ADD } E, F \}$



Run in parallel

④ Superscalar Microprocessor:

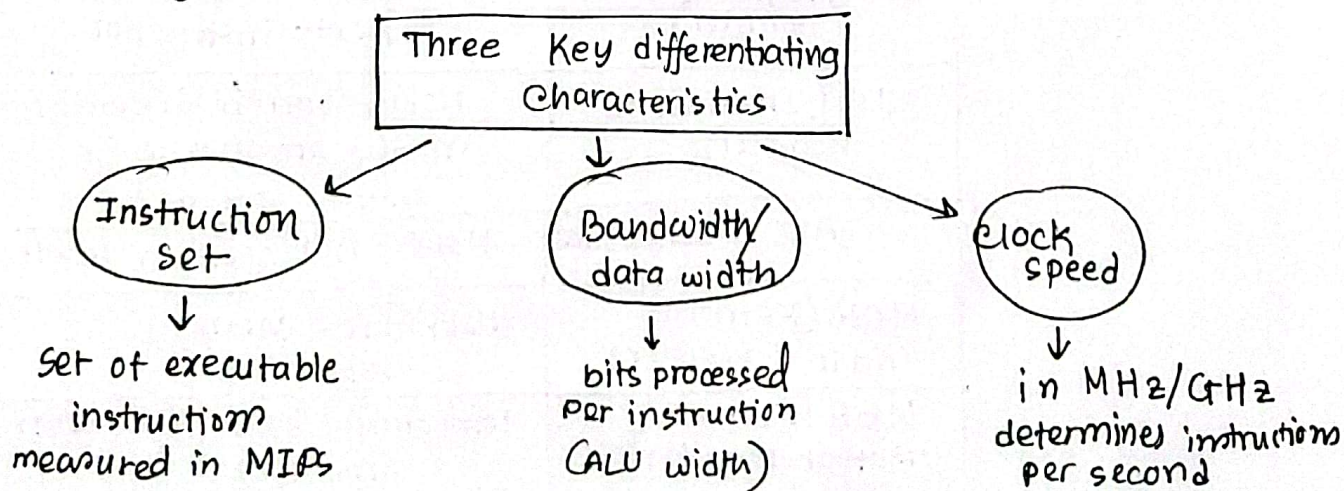
- Executes more than one instruction per clock cycle
- has multiple execution units inside processor.
- allows several instruction to run at the same time

⑤ ASIC → Application specific Integrated Circuit

- Designed for specific task, not general purpose
- used in smartphone, routers, crypto mining, TVs

★ Characteristics of Microprocessor:

- ① Integrates CPU functions on a single IC
- ② Performs arithmetic and logic using ALU
- ③ Can move data between locations
- ④ Has program counter for storing next instruction address
- ⑤ Computational capacity high
- ⑥ Performs fetch → decode → execute cycle
- ⑦ Needs external RAM, ROM, I/O for complete system
- ⑧ High building cost
- ⑨ High power consumption.



★ Hardware of a Microprocessor:

① Arithmetic logic unit (ALU)

Performs:

- Addition & Subtraction
- Multiplication and Division
- Logical tests
- Comparison
- Bit Shifting

② Control unit (CU)

Directs and Manages instruction execution

③ Registers:

~~Small storage location for quick data access~~

④ Buses (Data, Address, Control):

~~Transfer data and signals between components.~~

Three elements:

Decoder



decodes program instruction

Timer/Clock



synchronizes all processes

Control Logic Circuits



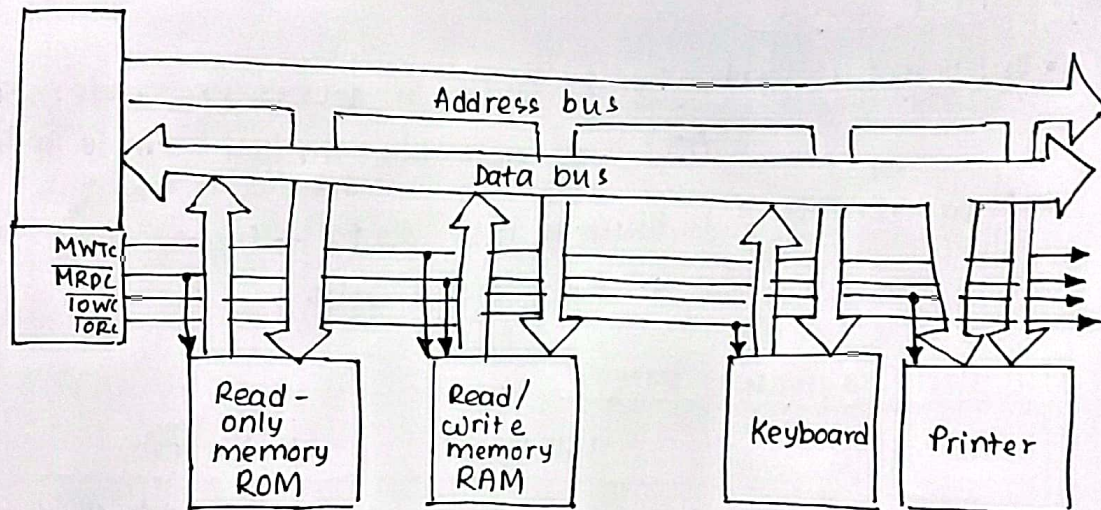
generates control signals

⑤ Register:

Register	=	function
PC (Program counter)		Holds address of next instruction
IR (Instruction Register)		Holds current instruction in the processor while being decoded
ACC (Accumulator)		Holds ALU operation results
MAR (Memory Address Register)		Holds stores Memory addresses
MBR (Memory Buffer Register)		Temporarily holds data from memory

Register	function
Flag /status Register	Holds 1-bit status register flags
General purpose register	Quick temporary data storage

★ System Bus



Three types of buses

Bus	Direction	Function
Data Bus	Bi-Directional	Transfers data between MP, memory & I/O
Address Bus	Unidirectional	Request specific memory I/O location
Control Bus	Unidirectional	Selects memory I/O and controls read/write

control bus signals:

- MRDC → Memory Read Control
- IORC → I/O Read control
- MWTC → Memory write control
- IOWC → I/O write control

Function of Buses:

- ① Data sharing → transfers data between peripherals
- ② Addressing → sends data to specific memory locations
- ③ Power supply → supplies power to peripherals
- ④ Timing → provides clock signal to synchronize peripherals

★ Expansion Buses:

- Integrated Motherboards: Built in devices (keyboard, mouse, LAN) - cheaper but limited, simple design
Expensive to repair
- Expansion Cards: Plugged into slots - higher performance
(GPU, sound cards)

Common Expansion buses:

Type	Full Name	Purpose
PCI	Peripheral component Interconnect	Connects peripherals to motherboard (Intel early 1990s)
PCMCIA	Personal computer Memory card Industry Association	For laptops - adds modem, storage, network cards
AGP	Accelerated Graphics port	Dedicate graphics card connection (Intel 1997)
SCSI	Small computer system Interphase	Connects multiple high speed devices (HDDs, scanners)
pc card	Based on PCMCIA standard	Small cards for laptop

★ How a microprocessor works — Fetch - Decode - Execute cycle

Fetch:

- ① PC copies next instruction address $\rightarrow$ MAR
- ② MAR places address on address bus
- ③ MAR triggers a read signal $\rightarrow$ Ram places instruction on DATA bus.
- ④ Instruction loaded into MBR (Memory Buffer Register)
- ⑤ MBR copies instruction to Instruction Register (IR)

Decode:

- control unit examines the instruction in IR
- The decode unit interprets it. (e.g., recognizes "ADD")
- Prepares the CPU to carry out the instruction.

Execute:

- ALU carries out the instruction
- Result stored in appropriate register.

★ Memory Addressing

- Each byte in memory has a unique address. (location/number)
- Formula: $N = \log_2(M)$

where N = address bits

M = memory size in bytes

Memory Size	Address Lines Needed
64 KB	16 bits
1 MB	20 bits
4 GB	32 bits

- Lowest address : 00000H
- Highest address : FFFFAH

Intel Processor Generations and architecture:

Processor	Year	Architecture	clock Speed	Address bus	Key Feature
80186	1982	16-bits	6-25MHz	20 bit (1MB)	Embedded systems, Integrated components
80286	1982	16-bit	6-25MHz	24 bit (16MB)	Protected Mode (Multitasking)
80386	1985	32 bit	12-40MHz	32 bit (4GB)	First full 32-bit, MMU, virtual 8086 mode
80486	1989	32 bit	up to 50 MHz	32 bit	Integrate FPU, faster execution
Pentium	1993	64-bit	60-133 MHz	—	Superscalar, 110 MIPS, 16 KB cache

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